

leetcode.com/problems/insert-interval/description/

Useful test cases



Description

Description



Note

Note



Editorial

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Solutions

Solutions



Submissions

Submissions

</>

Code

Code



Testcase

Testcase

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Test Result

Test Result



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57. Insert Interval

Solved ●

Medium



Topics

Companies



Hint

You are given an array of non-overlapping intervals `intervals` where `intervals[i] = [starti, endi]` represent the start and the end of the i^{th} interval and `intervals` is sorted in ascending order by `starti`. You are also given an interval `newInterval = [start, end]` that represents the start and end of another interval.

Insert `newInterval` into `intervals` such that `intervals` is still sorted in ascending order by `starti` and `intervals` still does not have any overlapping intervals (merge overlapping intervals if necessary).

Return intervals *after the insertion*.

Note that you don't need to modify intervals in-place. You can make a new array and return it.

Example 1:

Input: intervals = [[1,3],[6,9]], newInterval = [2,5]

Output: [[1,5],[6,9]]

Example 2:

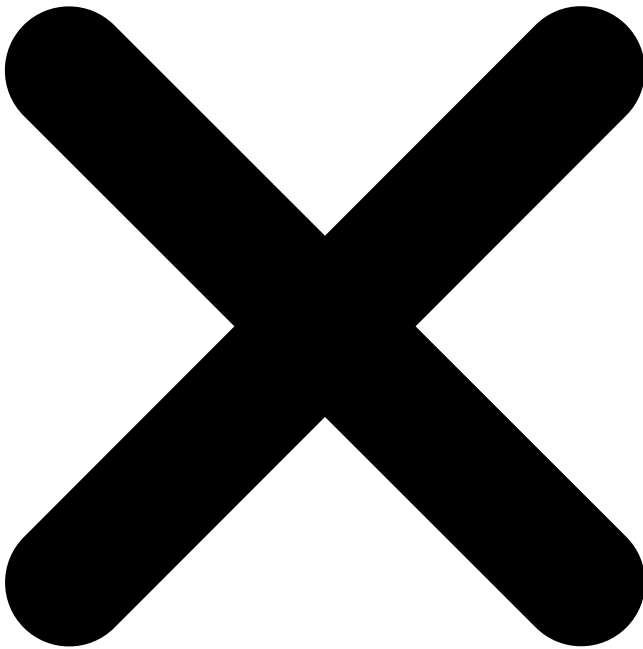
Input: intervals = [[1,2],[3,5],[6,7],[8,10],[12,16]], newInterval = [4,8]

Output: [[1,2],[3,10],[12,16]]

Explanation: Because the new interval [4,8] overlaps with [3,5],[6,7],[8,10].

Constraints:

- $0 \leq \text{intervals.length} \leq 10^4$
- $\text{intervals}[i].\text{length} == 2$
- $0 \leq \text{start}_i \leq \text{end}_i \leq 10^5$
- intervals is sorted by start_i in **ascending** order.
- $\text{newInterval.length} == 2$
- $0 \leq \text{start} \leq \text{end} \leq 10^5$



Seen this question in a real interview before?

1/6

Yes

No

Accepted

1,802,926/4M

Acceptance Rate

45.3%



Topics



Array



Companies



Hint 1



Intervals Array is sorted. Can you use Binary Search to find the correct position to insert the new Interval.?



Hint 2



Can you try merging the overlapping intervals while inserting the new interval?



Hint 3



This can be done by comparing the end of the last interval with the start of the new interval and vice versa.



Discussion (434)



 Discussion Rules

X

1. Please don't post **any solutions** in this discussion.
2. The problem discussion is for asking questions about the problem or for sharing tips - anything except for solutions.
3. If you'd like to share your solution for feedback and ideas, please head to the solutions tab and post it there.

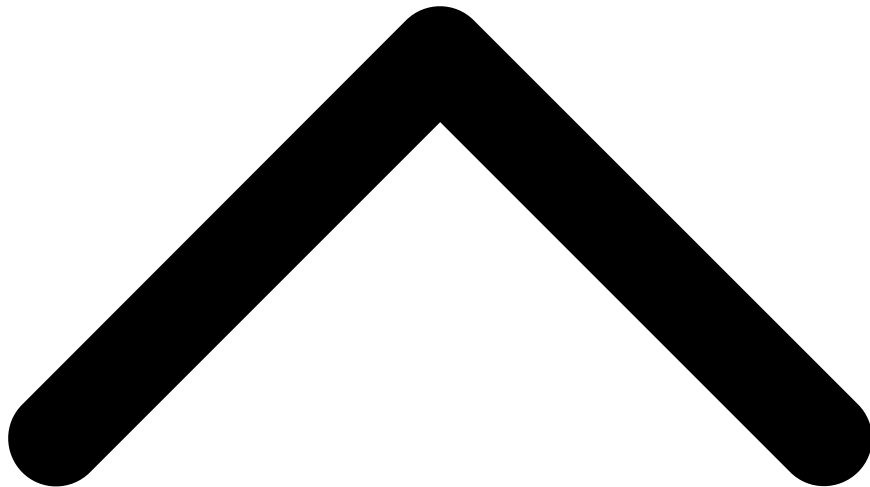


Taswell



Jan 15, 2023

this problem feels like its sole purpose of existing is to disgust you.



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1.1K

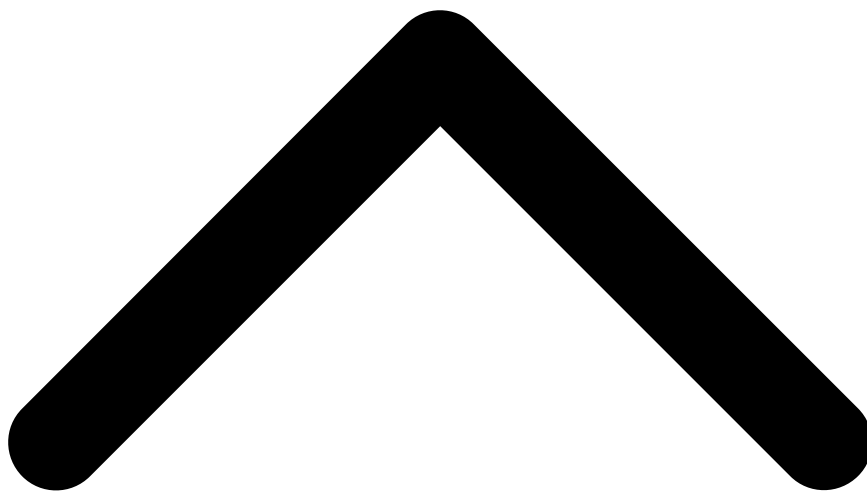


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with each failed test case and after adding another `if` statement, I realize that there is a simpler solution, but I still persist in branching in the hope that next attempt will definitely be accepted..



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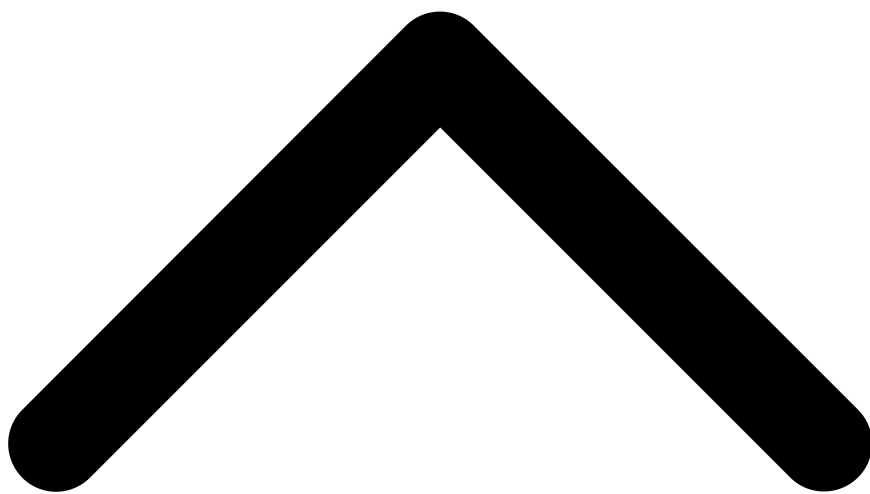


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Whenever the times comes when I start developing arrogance about the multiple data structures + algos I know, I will return back to this problem and it will humble me down by telling me **how a simple Array problem with simple statement can test our coding abilities!!**



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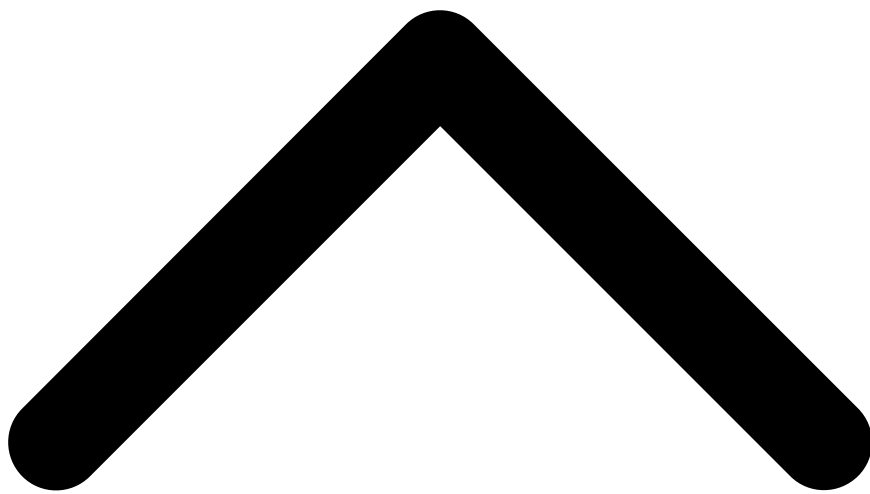


Julien



Jan 17, 2023

I am so relieved looking at the comments to see that I was not the only one being completely thrown off by how easy the problem sounds and what a nightmare it turned out to be! Every failed test case took away a piece of my soul. There's very little left of it but after several hours of pure frustration, I at least got a horrible frankenstein monster of a solution, with awful if statements pieced together. I need to work read up in the solution and learn but that will be for another (distant) day



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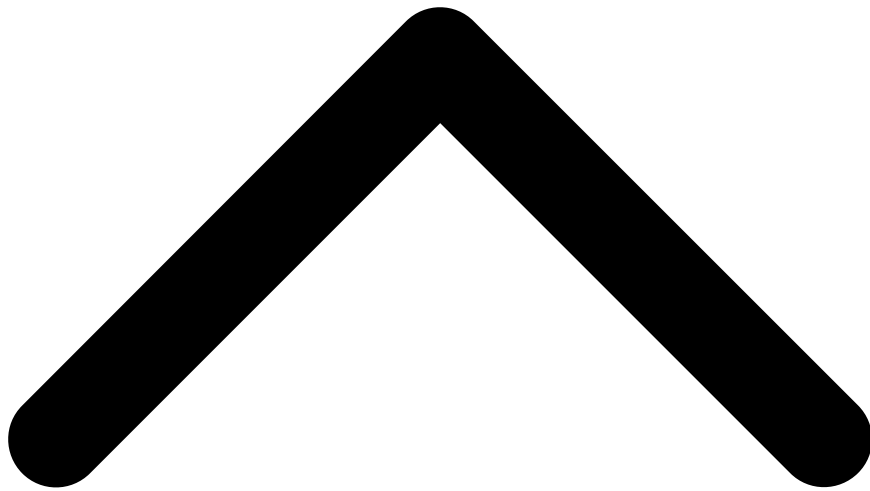
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Adi



Jan 16, 2023

Congrats to the people who made it through the forest!!!



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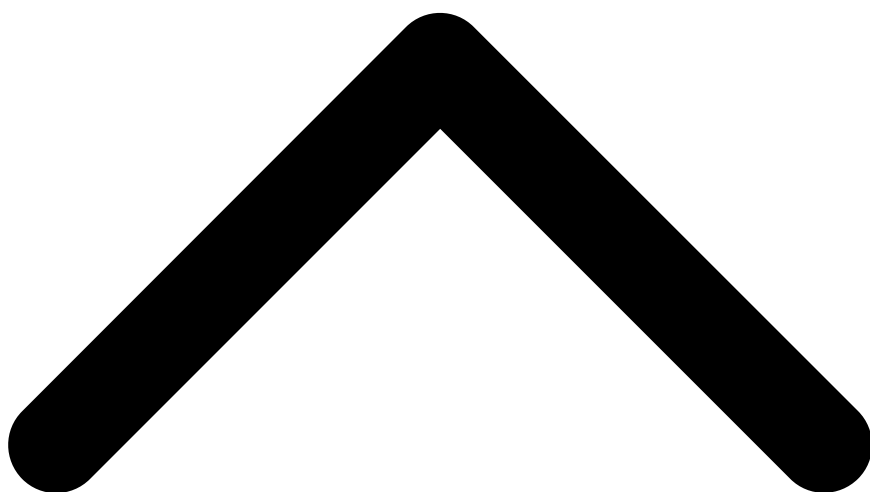


Alec Kerrigan



Jan 15, 2023

I receive this problem from startups all the time, I'm not sure why they like it, but really work at it!



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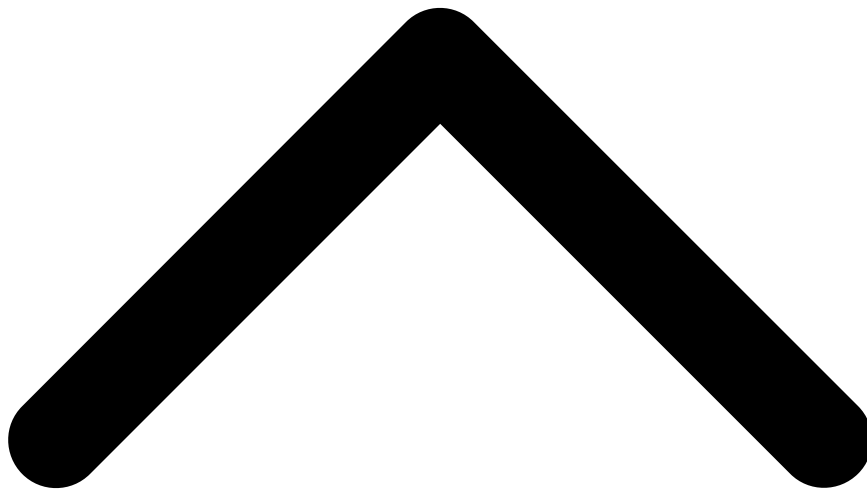


Madhav Sarpal



Jan 15, 2023

Finally a stop to tree problems! 🐵🐵



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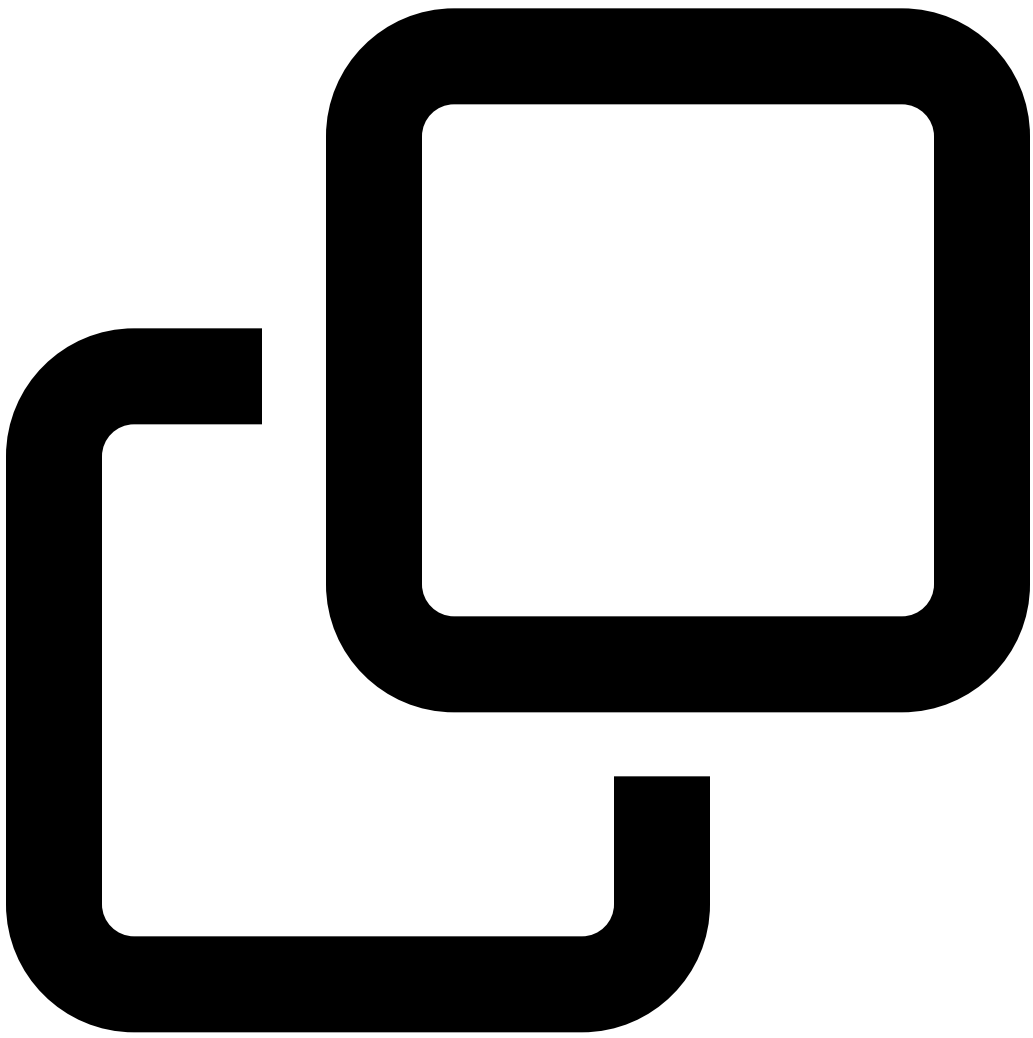


Ogabek



Mar 16, 2024

```
[ ]  
[2,5]  
[[3,5],[8,10]]  
[1,2]  
[[1,3],[6,9]]  
[11,12]  
[[1,3],[6,9]]  
[2,5]  
[[1,2],[3,5],[8,10]]  
[2,9]  
[[1,3],[6,9]]  
[2,7]  
[[1,3],[6,9]]  
[4,7]  
[[1,3],[6,9]]  
[3,6]
```



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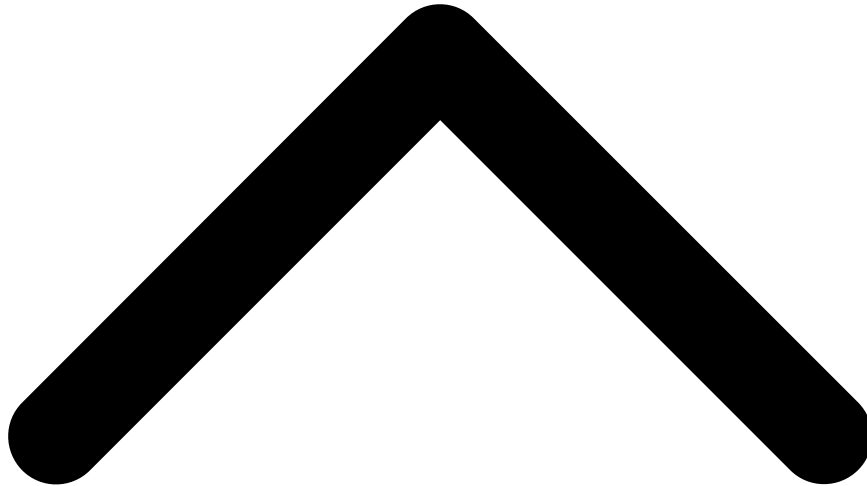


todamoon



May 02, 2019

It feels to me only adding unnecessary coding complexity without adding interesting challenge to this question. Any one sees the point of such change?



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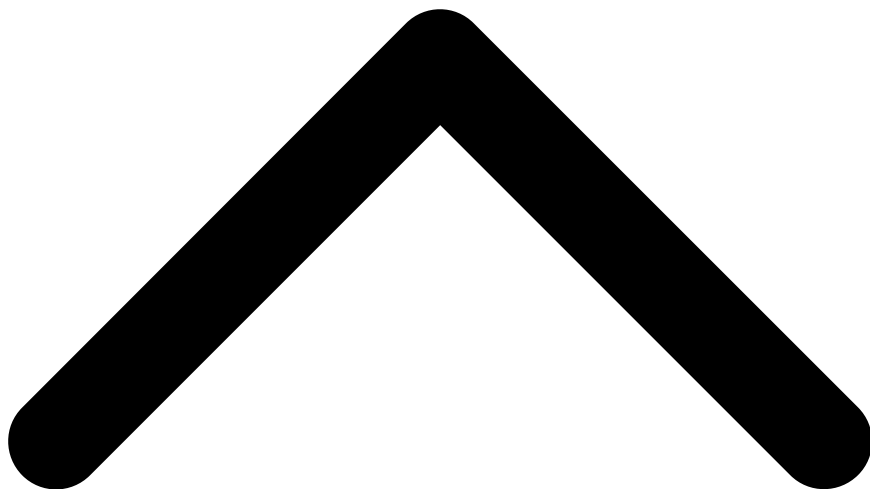
Benjamin Lee



Apr 19, 2023

"Return intervals after the insertion."

Can the question be rephrased? This sentence makes it seems like the code is asking for us to modify the input intervals array, but the problem actually can be solved by creating a new solution array.



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Feedback



42





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```
import java.util.ArrayList;
```

```
public class Solution {
```

```
    public int[][] insert(int[][] intervals, int[] newInterval) {
```

```
        //creates and initializes variables
```

```
        ArrayList<int[]> result = new ArrayList<>();
```

```
        for(int i = 0; i < intervals.length; ++i) {
```

```
            if(newInterval[1] < intervals[i][0]) {
```

```
                result.add(newInterval);
```

```
                for(int j = i; j < intervals.length; ++j)
```

```
                    result.add(intervals[j]);
```

```
                return result.toArray(new int[0][]);
```

```
            } else if(newInterval[0] > intervals[i][1]) {
```

```
                result.add(intervals[i]);
```



Saved

Ln 1, Col 1

intervals =

[[1,3],[6,9]]

newInterval =

[2,5]

9

1

2

3

4

>

[[1,3],[6,9]]

[2,5]

[[1,2],[3,5],[6,7],[8,10],[12,16]]

[4,8]

</>

Source



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